

Which of the following options has the same meaning as a 'compression wave'?

- A A vibration wave
- B An elastic wave
- C A longitudinal wave
- D A transverse wave

Which of the following statements best describes a wave?

- A A wave transfers energy to vibrating particles that then return the energy to the source.
- B A wave transfers energy into a material by vibrating the particles in the material.
- C A wave transmits energy through a material by moving particles from one place to another.
- D A wave transmits energy through a material with no overall movement of matter.

Which of the following statements best describes the motion of particles in a transverse wave?

- A The particles vibrate parallel to the direction of energy transfer.
- B The particles vibrate at right angles to the direction of energy transfer.
- C The particles gradually spread out parallel to the direction of energy transfer.
- D The particles gradually spread out at right angles to the direction of energy transfer.

Which of the following statements about sound waves travelling through a medium is **not** true?

- A 100% of the energy in the sound passes completely through the medium.
- B Sound travels more efficiently through elastic materials.
- C Particles of the medium vibrate but do not change their average positions.
- D Particles of the medium transfer energy to neighbouring particles.

Which of the following lists correctly arranges the different types of materials in order of increasing speed of sound (i.e. from slowest to fastest)?

- A Solids; liquids; gases
- B Solids; gases; liquids
- C Liquids; solids; gases
- D Gases; liquids; solids

Which of the following statements best describes why the speed of sound in steel is greater than the speed of sound in air?

- A The particles in steel are more closely bound together.
- B The particles in steel are heavier.
- C The particles in steel are larger.
- D The particles in steel vibrate at higher frequencies.

Which of the following statements best describes the motion of particles in a transverse wave?

- A The particles vibrate parallel to the direction of energy transfer.
- B The particles vibrate at right angles to the direction of energy transfer.
- C The particles gradually spread out parallel to the direction of energy transfer.
- D The particles gradually spread out at right angles to the direction of energy transfer.

Which of the following explanations best describes the relation between a 'ray' and a 'beam'?

- A Rays travel in straight lines; beams are formed when rays are deflected.
- B A beam is what is seen when rays are scattered.
- C A beam shows the path of light; a ray is a bundle of beams.
- D A ray shows the path of light; a beam is a bundle of rays.

Plane mirrors can be used to reflect light rays from an object and form an image. Which of the following options best describes the relation between the light rays and the image?

- A The image is where the reflected rays converge together.
- B The image is where all the reflected rays appear to come from.
- C The image is where all the reflected rays come from.
- D The image is on the mirror surface, opposite the object.

Which of the following statements best describes the image formed in a plane mirror?

- A The same distance behind the mirror as the object is in front; enlarged.
- B The same distance behind the mirror as the object is in front; smaller.
- C The same distance behind the mirror as the object is in front; upright.
- D The same distance behind the mirror as the object is in front; upside down.

Which of the following statements best describes how the eye changes when looking at close-up objects?

- A The pupil expands to allow more light in.
- B The lens moves forward in the eyeball.
- C The lens is stretched thin.
- D The lens bulges or thickens.

Which of the following choices best describes the operation of an endoscope used to inspect a baby in the womb?

- A Light travels, by reflection in mirrors, along one cable of optical fibres into the womb. The light leaves the body along a second such cable and is imaged by a lens.
- B Light travels, by total internal reflection, along one cable of optical fibres into the womb. The light leaves the body along a second such cable and is imaged by a lens.
- C Light travels, by refraction, along one cable of optical fibres into the womb. The light leaves the body along a second cable and is imaged by a lens.
- D Light travels, by total internal reflection, along a single cable of optical fibres, firstly into the womb and then out of the body. A lens finally images the light.

Which of the following statements is **not** a property of the lens shown below?



- A It causes parallel light rays to converge.
- B It causes parallel light rays to diverge.
- C It is a biconvex lens.
- D It has a real focal point.